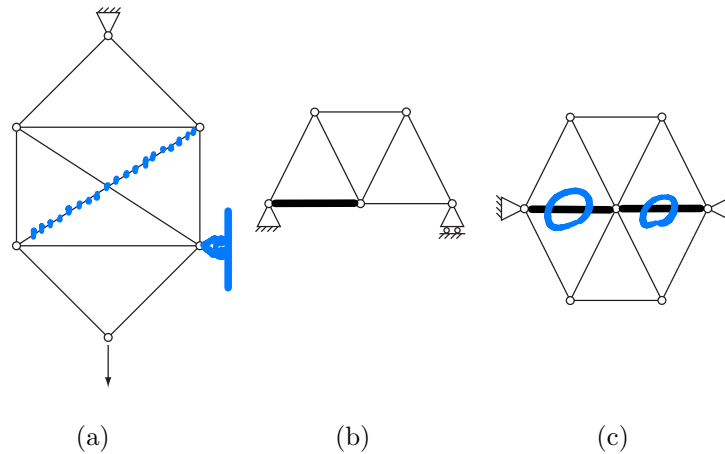


AERO40008 – Structures

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Static and Kinematic Determinacy

(1.) For each of the three pin-jointed frameworks below, determine whether the structure is a mechanism, redundant, or statically-determinate. If the structure is redundant or a mechanism, sketch how the structure could be modified to make it statically-determinate.



$$M = b + r - 2j$$

$$(a) \quad M = 10 + 2 - 2 \times 6 = 0$$

By inspection, statically-determine Mechanism

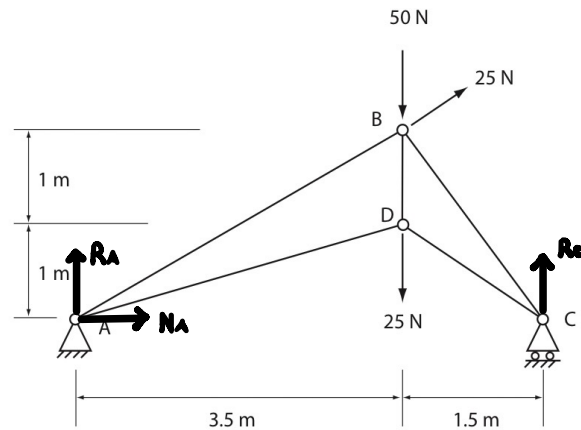
$$(b) \quad M = 6 + 3 - 2 \times 5 = -1 \quad \therefore \text{Mechanism}$$

$$(c) \quad M = 10 + 4 - 2 \times 7 = 0$$

By inspection, mechanism statically-determine

Method of Sections

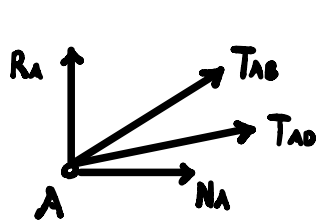
(2.) Determine the force in member BD under the loading shown.



$$(\uparrow) R_A + R_C = 50\text{ N} + 25\text{ N} - 25\text{ N} \times \sin\left(\tan^{-1}\left(\frac{2\text{ m}}{3.5\text{ m}}\right)\right) = 62.597\text{ N}$$

$$(\rightarrow) N_A = -25\text{ N} \times \cos\left(\tan^{-1}\left(\frac{2\text{ m}}{3.5\text{ m}}\right)\right) = -21.706\text{ N}$$

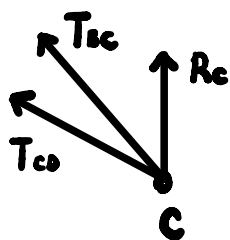
$$M(A), 5\text{ m} \times R_C = 3.5\text{ m} \times (50\text{ N} + 25\text{ N}) \quad \therefore R_C = 52.5\text{ N} \quad \therefore R_A = 10.097\text{ N}$$



$$(\rightarrow) T_{AB} \cos\left(\tan^{-1}\left(\frac{4}{7}\right)\right) + T_{AD} \cos\left(\tan^{-1}\left(\frac{2}{7}\right)\right) + N_A = 0$$

$$(\uparrow) T_{AB} \sin\left(\tan^{-1}\left(\frac{4}{7}\right)\right) + T_{AD} \sin\left(\tan^{-1}\left(\frac{2}{7}\right)\right) + R_A = 0$$

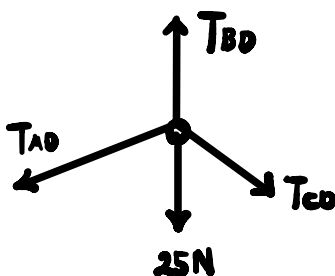
$$\therefore T_{AD} = 81.903\text{ N} \quad \therefore T_{AB} = -65.702\text{ N}$$



$$(\rightarrow) T_{BC} \cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right) + T_{CD} \cos\left(\tan^{-1}\left(\frac{2}{3}\right)\right) = 0$$

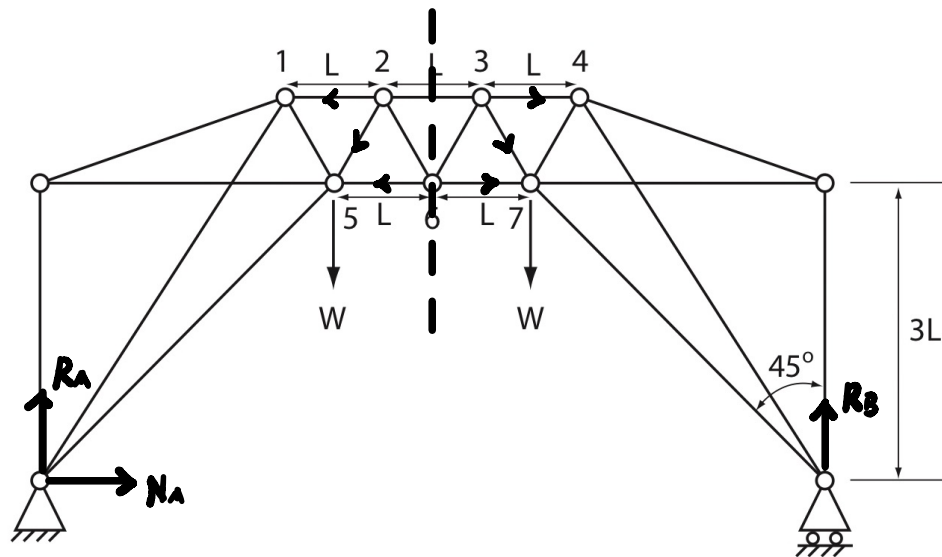
$$(\uparrow) T_{BC} \sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right) + T_{CD} \sin\left(\tan^{-1}\left(\frac{2}{3}\right)\right) + R_C = 0$$

$$\therefore T_{CD} = 94.616\text{ N} \quad \therefore T_{BC} = -131.25\text{ N}$$



$$\begin{aligned} (\uparrow) T_{BD} &= T_{AD} \times \sin\left(\tan^{-1}\left(\frac{2}{7}\right)\right) + T_{CD} \times \sin\left(\tan^{-1}\left(\frac{2}{3}\right)\right) \\ &\quad + 25\text{ N} \\ &= 100\text{ N} \end{aligned}$$

(3.) Determine the loads in all the bars which connect with nodes 2, 3, and 6 in the pin-jointed truss structure shown in the figure.



$$\therefore (\rightarrow) N_A = 0$$

Symmetry about the line shown in the diagram

$$\therefore T_{56} = T_{67}, T_{25} = T_{37}, T_{12} = T_{34}$$

$$R_A = R_B = W$$

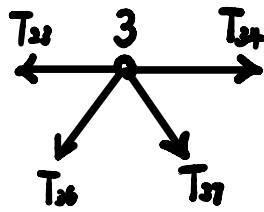
For method of sections :

$$M(3), \frac{\sqrt{3}}{2}L \cdot T_{67} + (3L + \frac{1}{2}L)R_B = \frac{1}{2}L \cdot W \quad \therefore T_{67} = 2\sqrt{3}W$$

$$M(7), \frac{\sqrt{3}}{2}L \cdot T_{34} = 3L \cdot R_B \quad \therefore T_{34} = -2\sqrt{3}W$$

$$M(6), \frac{\sqrt{3}}{2}L \cdot T_{37} + \frac{\sqrt{3}}{2}L \cdot T_{34} + L \cdot W = (3L + L)R_B$$

$$\therefore T_{37} = 0$$



$$(\rightarrow) T_{23} + \frac{1}{2}T_{36} = T_{34} + \frac{1}{2}T_{37} = -2\sqrt{3}W$$

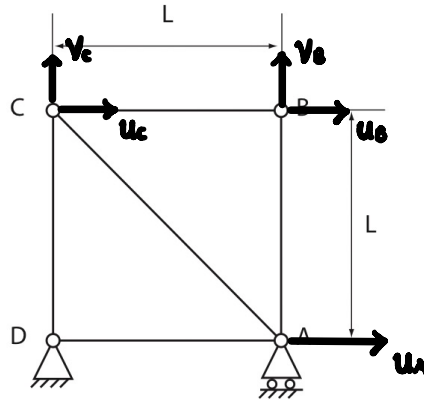
$$(\uparrow) \frac{\sqrt{3}}{2}T_{36} + \frac{\sqrt{3}}{2}T_{37} = 0 \quad \therefore T_{36} = 0 \quad \therefore T_{23} = -2\sqrt{3}W$$

$$T_{23} = -2\sqrt{3}W$$

$$T_{26} = T_{36} = 0$$

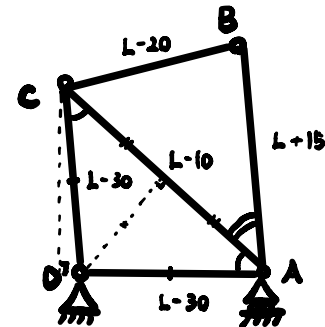
Direct Displacement Method

(4.) The members in the structure shown undergo the following extensions: AB +15 mm, BC -20 mm, AC -10 mm, AD -30 mm, CD -30 mm. Find the displacements of joints A, B, and C, and sketch the deformed shape.



$$M = b + r - 2j = 5 + 3 - 2 \times 4 = 0$$

By inspection, statically determinate



$$\therefore u_A = -30 \text{ mm}$$

$$u_C = \left[\frac{L-10}{L-30} \cdot \frac{1}{2}(L-10) - L \right] \text{ mm} = \left(\frac{-L^2 + 40L + 100}{2L-60} \right) \text{ mm}$$

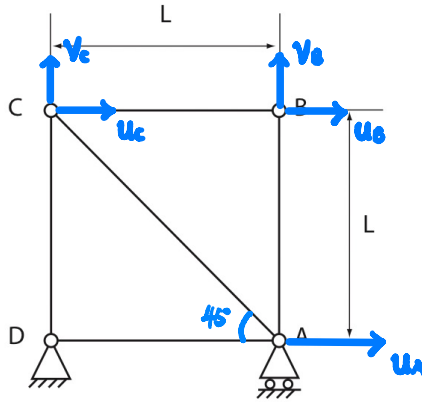
$$v_C = \left[\frac{L-10}{L-30} \sqrt{(L-30)^2 - \left(\frac{1}{2}L-5\right)^2} - L \right] \text{ mm}$$

$$u_B = \left[(L+15) \sin \left(\cos^{-1} \left(\frac{L-10}{2L-60} \right) + \cos^{-1} \left(\frac{(L-10)^2 + (L+15)^2 - (L-20)^2}{2(L-10)(L+15)} \right) \right) - L \right] \text{ mm}$$

$$v_B = \left[(L+15) \cos \left(\cos^{-1} \left(\frac{L-10}{2L-60} \right) + \cos^{-1} \left(\frac{(L-10)^2 + (L+15)^2 - (L-20)^2}{2(L-10)(L+15)} \right) \right) - L \right] \text{ mm}$$

Direct Displacement Method

(4.) The members in the structure shown undergo the following extensions: AB +15 mm, BC -20 mm, AC -10 mm, AD -30 mm, CD -30 mm. Find the displacements of joints A, B, and C, and sketch the deformed shape.



$$e_{AB} = v_b$$

$$e_{CD} = v_c$$

$$e_{AD} = u_a$$

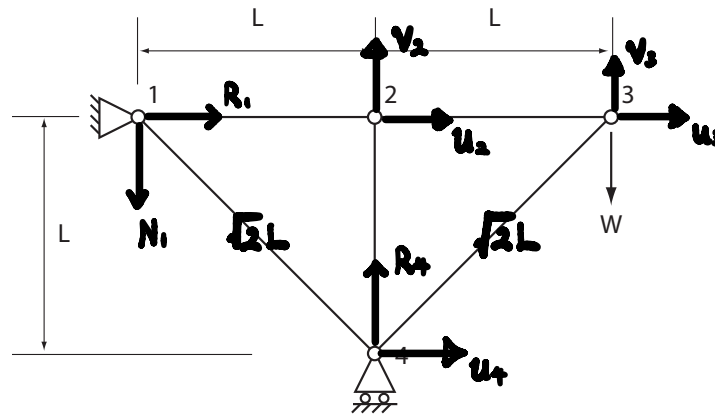
$$e_{BC} = u_b - u_c$$

$$e_{AC} = \frac{\sqrt{2}}{2}(u_a - u_c) + \frac{\sqrt{2}}{2}v_c$$

$$\therefore u_b = -66 \text{ mm}$$

$$u_c = -46 \text{ mm}$$

(5.) In the structure shown in the figure, all members have cross-section area A , and Young's modulus E . Determine the joint displacements under the loading shown, and sketch the deformed shape.



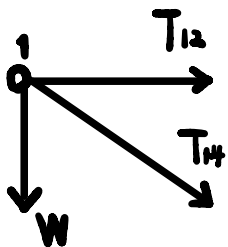
$$M = b + r - 2j = 5 + 3 - 2 \times 4 = 0$$

By inspection, statically determinate

M. Santer

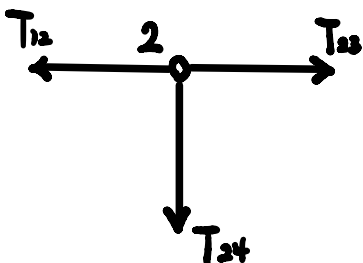
$$(\rightarrow) R_1 = 0 \quad (\uparrow) N_1 + W = R_4$$

$$M(1), L \cdot R_4 = (L + L) \cdot W \quad \therefore R_4 = 2W \quad \therefore N_1 = W$$



$$(\rightarrow) T_{12} + \frac{\sqrt{2}}{2} T_{14} = 0 \quad (\uparrow) \frac{\sqrt{2}}{2} T_{14} + W = 0$$

$$\therefore T_{14} = -\sqrt{2}W \quad \therefore T_{12} = W$$



$$(\rightarrow) T_{12} = T_{23} \quad (\uparrow) T_{24} = 0$$

By symmetry, $T_{34} = T_{14} = -\sqrt{2}W$

$$e_{12} = \frac{T_{12}L}{AE} = \frac{WL}{AE} = e_{23}$$

$$e_{14} = \frac{T_{13}L}{AE} = \frac{-\sqrt{2}W\sqrt{2}L}{AE} = -\frac{2WL}{AE} = e_{34}$$

$$e_{24} = 0$$

$$\therefore e_{12} = u_2, e_{23} = u_3 - u_2, e_{34} = v_2,$$

$$e_{14} = \frac{\sqrt{2}}{2}u_4, e_{24} = \frac{\sqrt{2}}{2}(u_3 - u_4) + \frac{\sqrt{2}}{2}v_3$$

$$\therefore u_2 = \frac{WL}{AE}, u_3 = \frac{2WL}{AE}, v_2 = 0$$

$$u_4 = -\frac{2\sqrt{2}WL}{AE}, v_3 = (-2-4\sqrt{2})\frac{WL}{AE}$$